LESSON 13 · MISSION GUIDE

Before the Throne

Charts people can read, and Excel files people can open.

From **Queen Aldara of Thornmere**
To **Jen, Keeper of the Guild Ledger**
By raven **at dusk, on the last day of the week**

Keeper, Quartermistress Pell has shown us what you have been doing with the "pandas", and Orrin tells us you now speak to the Vault beneath the guild hall in its own tongue. Most impressive. We understood almost half of it.

The Privy Council sits in a fortnight. The Privy Council does not read tables. It reads *pictures*, ideally blue ones, and then asks for "the workbook" to fiddle with in the carriage home. Dalvric once drew the pictures by hand, by candlelight, for a whole moon, which may be why he sailed west. Can your Python draw them and make the workbook? With bold headings? No pressure. (Some pressure.) — A.

What you'll be able to do after today

- [ ] Draw a **bar chart** and a **histogram** from a DataFrame with matplotlib, and save them as PNG files
- [ ] Put several series **side by side** (success, fled, failed per difficulty) and label everything properly
- [ ] Write a DataFrame to an **Excel workbook**, and several tables to several **sheets** of one workbook
- [ ] Format a workbook with openpyxl: bold coloured headers, `#,##0.00 "cr"` number formats, column widths, frozen panes
- [ ] Drop a chart image into a worksheet

Why this matters at work

- **The last mile is the one people see.** Nobody on the Privy Council will run your script. They'll open a PNG and an xlsx in the carriage home. If that looks right, the whole project looks right.
- **Formatting by code is formatting forever.** Set the header style once in Python and every season's report has it. No more "can you make it look like last season's".

Four big ideas

1. A chart is a file	In a script there is no window. `plt.savefig("x.png")` writes the picture; you open it like any file. `matplotlib.use("Agg")` at the top tells matplotlib not to look for a screen.
2. Build, then decorate	Order of a chart: `plt.figure()` → draw (`bar`, `hist`) → title, labels, legend → `tight_layout()` → `savefig()`. Same every time.
3. pandas writes Excel	`df.to_excel(path, sheet_name=...)` for one sheet; `with pd.ExcelWriter(path) as writer:` for several. Needs openpyxl installed, which is why it's in `requirements.txt`.
4. openpyxl edits the workbook	pandas writes values. For looks, re-open with `load_workbook`, walk the cells, set `font`, `fill`, `number_format`, widths, then `save`.

Things that will bite you

- Calling `plt.show()` in a script: it either hangs or does nothing. Save instead.
- Drawing a second chart without `plt.figure()`: both charts land on the same picture.
- Excel number formats are text: `'#,##0.00 "cr"'`. Exactly the string you'd type in Format Cells (the inner quotes are part of it).
- openpyxl colours have no `#`: `"1F3A5F"`. matplotlib colours do: `"#1f3a5f"`. And both spell it `color`. Yes, really.

How today works

15 MIN
Lecture
Lars draws two charts and writes one workbook, live

60 MIN
Exercise sheet
Seven ravens: three charts, four workbooks

15 MIN
Pages from the Grimoire, page 13
The Council pack in one click. The Queen clicked.

The ravens

#	FROM	RAVEN	YOU'LL USE
1	Queen Aldara	Your first chart	`plt.bar savefig`
2	Captain Siv	What does a quest pay?	`plt.hist axvline legend`
3	Quartermistress Pell	Side by side	`pivot_table np.arange` grouped bars
4	Orrin	Back to Excel	`to_excel`
5	Quartermistress Pell	One workbook, several sheets	`pd.ExcelWriter merge`
6	Queen Aldara	Make it pretty	`load_workbook Font PatternFill number_format`
7 ⭐	Queen Aldara	A chart inside the workbook	`barh add_image`

You're done when...

- [ ] `my_work/output/` holds four PNGs and four workbooks, and you've opened every one of them
- [ ] Every script prints what the sheet says, and Dalvric's Council pack names the right three heroes
- [ ] Everything is pushed to `lesson-13`, with your favourite chart attached to the pull request



Next lesson: The Charter. The finale. Two sessions, one report: the Charter Report for Queen Aldara, straight from the Vault and the ledgers, with a chart, a workbook, and one sentence the Queen can say out loud at Midsummer. Everything you've learned, in one project folder.